BDS-SMC2 UAV Rescue System

IEEE Publication Plan — Two Papers

Letsoalo Maile | GNSS & Satellite Communication Dissertation

June 2026

1. Publication Strategy

The five research gaps addressed in this dissertation form two distinct contribution clusters. Splitting them into two papers maximises impact, avoids desk rejection for scope, and matches the data requirements of each venue.

Paper	*Gaps*	*Target Venue*	*Core Claim*
Paper 1	Gap 2 + Gap 3	IEEE Transactions on Aerospace & Electronic Systems (TAES)	BDS-3 SMC latency characterisation + environmental reliability for UAV rescue
Paper 2	Gap 1 + Gap 6	IEEE Communications Letters	Payload encoding efficiency: Binary vs Huffman for BDS-3 SMC telemetry

2. Paper 1 — IEEE TAES (Gap 2 + Gap 3)

Field	*Detail*
Full title	End-to-End Latency and Environmental Reliability of BeiDou-3 Short Message Communication for UAV Search-and-Rescue
Venue	IEEE Transactions on Aerospace and Electronic Systems (TAES)
Impact factor	5.1 | Q1 journal | Target length: 14 pages IEEE double-column
Submission	After 2-week experiment window — target August 2026
Gaps covered	Gap 2 (latency across sessions) + Gap 3 (4 environments x 3 locations)

2.1 Data Requirements (what you must collect)

Gap	*Requirement*	*Minimum*	*Target*	*Status*
Gap 2	TX latency across 3 time-of-day sessions	70 TX	100 TX	30 collected (baseline)
Gap 3	Success rate: 4 environments x 3 locations	20 TX/env	20 TX x 3 locations = 60/env	0 — hardware needed

2.2 Paper Structure (8 sections)

Section	*Title*	*Content*	*Write when*
I	Introduction	UAV rescue problem, BDS-3 SMC opportunity, 5 contributions	Now

II	Background & Related Work	BDS-3 RDSS architecture, prior latency studies, gap table	Now
III	System Design	ESP32 firmware, wiring, AT command protocol, portal	Now
IV	Methodology	Experiment setup, session design, environment classification, statistical plan	Now
V	Gap 2 Results — Latency	Mean 2.58s, sigma 1.09s, ANOVA, CDF, UAV error model	After hardware
VI	Gap 3 Results — Environments	Success rates, 95% CI, chi-square, Fisher's exact, Bonferroni	After hardware
VII	Discussion	UAV update rate limit, environment ranking, comparison to LTE/LoRa	After hardware
VIII	Conclusion	Summary, limitations, future work	Last

2.3 Key Results to Report (from existing + new data)

Metric	*Value*	*Source*
Mean end-to-end TX latency	2582.9 ms	gap2_latency.csv (n=30)
Std deviation	1093.7 ms	gap2_latency.csv
Latency range	1070 – 4488 ms	gap2_latency.csv
Decode overhead	8.4 ms	gap2_latency.csv
UAV max error @ 10 m/s	25.8 m (mean), 44.9 m (P95)	Derived
ANOVA p-value	TBD after 3 sessions	gap2_analysis.py
Open sky success rate	TBD	gap3_field_test.csv
Indoor success rate	TBD (expected < 30%)	gap3_field_test.csv
Chi-square across envs	TBD (expected p < 0.001)	gap3_analysis.py

2.4 Task List — Paper 1

#	*Task*	*When*	*Tool*
1	Collect Gap 2: 70 more TX across 3 sessions	Hardware days 1-4	serial_logger.py
2	Collect Gap 3: 4 envs x 3 locations x 20 TX	Hardware days 2-5	field_test_logger.py
3	Run gap2_analysis.py — ANOVA + CDF + UAV error	After Day 5	gap2_analysis.py
4	Run gap3_analysis.py — chi-square + Fisher's + Bonferroni	After Day 5	gap3_analysis.py
5	Regenerate all figures (Gap 3 auto-generates)	After analysis	generate_figures.py
6	Write Sections I – IV (no new data needed)	Now / before hw	IEEE TAES template
7	Download IEEE TAES LaTeX/Word template	Now	ieeeauthorcenter.ieee.org

8	Compile 30 references in IEEE format	Now	Google Scholar / Zotero
9	Write Sections V – VI with real numbers	After analysis	Paper draft
10	Write Sections VII – VIII (discussion + conclusion)	After Sec V-VI	Paper draft
11	Supervisor review — send full draft	End of week 2	Email
12	Revise and submit to TAES	Target Aug 2026	IEEE Manuscript Central

3. Paper 2 — IEEE Communications Letters (Gap 1 + Gap 6)

Field	*Detail*
Full title	Payload Encoding Efficiency for BeiDou-3 Short Message Communication in UAV Rescue: Binary Outperforms Huffman
Venue	IEEE Communications Letters
Impact factor	4.1 | Q1 | Length: 5 pages IEEE two-column (Letters format)
Submission	Target October 2026 (after Paper 1 draft is complete)
Gaps covered	Gap 1 (ASCII vs Binary encoding), Gap 6 (full telemetry struct comparison: Binary vs Huffman)

3.1 Data Requirements (mostly already collected)

Gap	*What you have*	*Additional needed*	*Status*
Gap 1	264 bits (ASCII) vs 64 bits (Binary) — software verified	10 hardware TX each mode for validation	Hardware Day 1
Gap 6	ASCII=368b, Binary=128b (-65.2%), Huffman=184b (-50%)	10 hardware TX with Huffman mode	Hardware Day 1

3.2 Key Results to Report (all already confirmed)

Gap	*Metric*	*Value*
Gap 1	ASCII lat/lon payload	264 bits
Gap 1	Binary lat/lon payload	64 bits (–75.8%, 4.12x)
Gap 6	ASCII full telemetry (7 fields)	368 bits (baseline)
Gap 6	Binary struct packing	128 bits (–65.2%)
Gap 6	Dynamic Huffman coding	184 bits (–50.0%)
Gap 6	Counterintuitive finding	Binary BEATS Huffman for structured numerical data — contradicts compression theory

3.3 Task List — Paper 2

#	Task	When
1	Hardware validate Gap 1: 10 ASCII + 10 Binary TX	Hardware Day 1
2	Hardware validate Gap 6: 10 Huffman TX	Hardware Day 1
3	Write 5-page Letters draft (Intro, System, Results, Discussion, Conclusion)	After Paper 1 draft
4	Download IEEE Communications Letters template	Before writing
5	Highlight the Binary-beats-Huffman finding as the key novelty	Section III of letter
6	Compile 15 references (shorter than Paper 1)	Before writing
7	Supervisor review	Target Sep 2026
8	Submit to IEEE Communications Letters	Target Oct 2026

4. Combined Timeline

Period	Activity	Output
Week 1 (now)	Build scripts, scout locations, write Paper 1 Sec I-IV	Scripts on standby, paper skeleton
Week 2 (hardware)	5 hardware days: collect all Gap 2 + Gap 3 data	gap2_latency.csv + gap3_field_test.csv complete
Week 3	Run analysis, regenerate figures, write Sec V-VI	All figures, stats tables, results sections
Week 4	Write Sec VII-VIII, full paper review, supervisor	Paper 1 first draft
Month 2	Revise Paper 1, write Paper 2 (5 pages, faster)	Paper 1 submission-ready, Paper 2 draft
August 2026	Submit Paper 1 to IEEE TAES	Under review
October 2026	Submit Paper 2 to IEEE Communications Letters	Under review